Resolution-Driven Dwells in SAR Task Scheduling: Mode Composition Decides What Tighter Requirements Cost

Liu Shuhao

College of Intelligent Systems Science and Engineering, Harbin Engineering University, 145 Nantong Street, Nangang District, Harbin, 150001, China

Abstract

Agile Earth-observation satellites that steer a single antenna across stripmap, sliding-spotlight, and spotlight modes are usually scheduled under an exogenous, fixed imaging duration. For a sliding-spotlight platform that assumption fails measurably: planning with the fixed library dwell violates the transfer constraint in 20–29% of consecutive pairs on the dense families and lands 18–30 points below the endogenous dwell. This paper makes the dwell endogenous and carries the required resolution as a per-task hard constraint: the mode parameters follow the requirement, the delivered resolution equals it at the selected start, and the stripmap tier stays at its antenna floor. Served priority is the only objective; image-quality readings are reported, not optimised. A frozen nine-point demand grid scales every requirement from 0.8 to 2.0, and three shape criteria are frozen in advance and reported with their failures. Tightening is where coverage moves: the dense 100-task family drops 26.0 percentage points from the as-placed requirement to the next point down and 46.6 points to the tightest feasible point; the single-satellite family drops 46.0 points at $s = 0.9$ and becomes infeasible at $s = 0.8$; the 50-task family is nearly flat, within 0.27 points throughout. Loosening from the as-placed point recovers 4.15 percentage points for the 100-task family against its own saturation and 2.36 points for the 50-task family (0.93 points paired per seed). Which mode a task receives

Email address: liushuhao@hrbeu.edu.cn (Liu Shuhao)

decides what a requirement change costs; conclusions are scoped to requirement sensitivity under this fixed mode assignment.

Keywords: agile Earth-observation satellite scheduling, synthetic aperture radar, sliding spotlight, per-task resolution requirement, coverage cost of a tighter requirement, demand sensitivity

---

1. Introduction

1.1. Motivation

Agile Earth-observation satellites steered by a single antenna can operate their synthetic aperture radar (SAR) in a continuum of imaging modes: stripmap, sliding spotlight, and spotlight are one physical mode family parameterised by where the beam's virtual rotation centre sits. Which point of that continuum a task receives is not a free stylistic choice — it is fixed by the requested resolution — the requirement, as this paper calls it from here on — the scene, and the geometry of the pass, and it determines how long the satellite must dwell on the scene. Scheduling therefore cannot treat the imaging duration as an exogenous constant, as the classical agile-scheduling literature largely does (Long et al., 2024; Qin et al., 2024; Li et al., 2024): in sliding-spotlight operation the dwell is produced by the imaging physics, it changes from task to task, and it feeds back into how many tasks a pass can carry and at what image quality. In that literature the resolution requirement, when it appears at all, is an attribute of the request that is settled before the schedule is built: the observation duration is a task attribute fixed in advance (Chen et al., 2024; Qin et al., 2024), and on the SAR side the required resolution already fixes the synthetic-aperture time and steering span of a designed observation (Kim and Chang, 2020).

This paper studies that coupling on a single satellite in a regime of single-pass dense task clusters under true orbit geometry. Scheduling decisions select which tasks to serve and when to start imaging within each task's visibility windows; the mode and its parameters follow from the product requirement,

and the dwell, the maneuver, and the image quality follow from the mode and the window. The systematic study of scenario density as an experimental axis is outside the scope of this paper; the density ladder is fixed at registration (Section 5.2).

Two discipline risks shape the methodology as much as the physics does. First, a measured effect on coverage can be a property of the *search* rather than of the *scenario*: in this world the greedy anchor already attains the family maximum in 258 of the 260 sweep units, so the coverage response reported here is produced by the mode and parameter adaptation rather than by the optimiser's ability to find it. Second, quality statements are only as good as their measurement calibers: several formally plausible formulations (an ambiguity sum that includes the principal image, a range resolution that varies per task, a squint-dependent routing resolution) are registered as not used, and the paper says so where it matters.

1.2. Contributions

Following the registered narrative frame — registered here meaning fixed in the project's preregistration before drafting began — the paper's contributions are three: the platform that carries the requirement into the physics, the mode-dependence of what a requirement costs in coverage, and the discipline that keeps both claims testable:

1. **An enabling physics-to-scheduler platform** (the platform leg). A three-mode continuum formalisation (Section 3) that carries the mode choice, the scene-level dwell closures, the required resolution as a per-task hard constraint, and the maneuver-coupled feasibility predicates into one scheduling loop: the imaging mode is resolved from the product requirement, the mode parameters follow the requirement where the mode has them so that the delivered resolution equals the requirement at the selected start, and the dwell duration is a consequence of the requirement rather than a constant — under a frozen mode assignment (Section 3.5).

2. **The mode-dependence of what a requirement costs** (main result). Three scenario families swept over a frozen nine-point demand grid at a single search budget and under a frozen mode assignment: a floor step on the stripmap family, a non-monotone response on the 50-task family whose two dips are traced to the greedy anchor construction, tightening costs of 26.0–46.6 percentage points on the dense family against a 0.27-point plateau on the 50-task family, as-placed costs of 4.15 and 2.36 percentage points against each family's own saturation, and a fixed-dwell counterfactual that quantifies what the endogenous dwell buys (Section 6).
3. **A discipline that keeps the result falsifiable** (methodology). Shape criteria frozen before this scenario world's data were read and reported with their failures rather than re-tuned to the data; caliber labels that keep the two quality calibers apart and keep the fulfillment ratio a reading; and an explicit separation of the search face from the construction face — the anchor attains the best-found coverage in 258 of 260 units, so the coverage response is a construction result, not a search finding (Sections 5 and 6).

Throughout, the trade-off the study claims is the resolution-versus-dwell-time axis of the mode continuum; the antenna-design question of resolution against swath is a sensor-layer consequence that this simulation does not model, and it is treated as a limitation with future work (Section 7.1). The serial-bound reading of how many dwell-coupled tasks a pass carries is stated with a per-pass caliber and without a number, because its serial bound has not been recomputed in this scenario world; it is carried as a scope statement (Section 7.1) rather than as a result.

The move this paper makes is to take the resolution off the request and put it into the constraint set: the requirement is what the scheduler must satisfy task by task, the mode parameters are what carry it into the physics, and the coverage loss is what the paper measures. The claim is scoped to the literature surveyed below: the survey rests on the project's curated literature pool — a converted

and catalogued bibliography of the field plus an incremental paper-search sweep over the months preceding writing, whose nearest neighbours Section 2 engages directly. What the tested coupling in fact produced in this scenario world is reported in Section 6: the coverage–demand curves of the three families, the floor step and the two anchor-carried dips, the as-placed costs against each family's own saturation, and the demand-satisfaction count — the platform's value is that each of these readings is registered and reproducible.

1.3. Paper organisation

Section 2 positions the work along sliding-spotlight imaging, agile-satellite scheduling, and quality-aware scheduling. Section 3 formalises geometry, modes, dwell, the requirement as a per-task constraint, and the demand-tightening axis. Section 4 describes the solver. Section 5 presents the scenario world and the frozen analysis design; Section 6 reports the coverage–demand curves and the demand-satisfaction readings; Section 7 discusses mechanisms, literature contrasts, and the full limitation set; Section 8 concludes.

2. Related Work

This section positions the paper along three lines: sliding-spotlight SAR imaging (where the dwell physics lives), agile Earth-observation satellite scheduling (where the fixed-dwell assumption lives), and quality-aware scheduling (where image quality enters planning decisions).

2.1. Sliding-spotlight and spotlight SAR imaging

The sliding-spotlight family is standardly formulated as steering about a fixed virtual rotation centre: the TerraSAR-X extended chirp scaling processor is built on that geometry (Mittermayer et al., 2003), the baseband azimuth-scaling processor of Prats et al. handles both sliding spotlight and terrain observation by progressive scans (TOPS)(Prats et al., 2010); in sliding spotlight the rotation centre lies beyond the swath, in TOPS it lies behind the sensor. Kim et al. derive the attitude-maneuver and synthetic-aperture time formulas for squint staring

and sliding spotlight on the same geometry (Kim et al., 2021). Operational descriptions sometimes state the steering as a constant angular rate — Zhang et al.'s Gaofen-3 (GF-3) account does so explicitly and describes it as a small-angle simplification of the same geometry (Zhang et al., 2018) — and processing work that pushes to high squint is likewise explicit that the linearised steering assumption breaks there (Guo et al., 2011; Ren et al., 2025). This paper adopts the rotation-centre formulation throughout (Section 3.1); its scheduling windows reach squints where only the exact geometry is valid, which is why the two readings must not be mixed.

Two quantitative anchors from the imaging line matter to this study, and both are quoted as their own platforms' literature values. First, Kim et al.'s Table 2 defines a $\pm 12^\circ$ task envelope for its reference mission; the design envelopes registered in this paper — $\pm 45^\circ$ for stripmap/spotlight and $\pm 26.9^\circ$ (a registered conservative value, a design choice) for sliding spotlight — are separate fields and are never merged with the literature task domain (Kim et al., 2021). Second, Song et al. report for their sparse-imaging platform that the azimuth ambiguity-to-signal ratio is anti-correlated with the pulse repetition frequency (PRF) and degrades as the scan angle grows (by about 8.6 dB beyond 1° of scan for conventional sliding spotlight, and about 15.2 dB for the sparse variant), alongside a range-ambiguity reduction of about 10.7 dB for the sparse mode (Song et al., 2025); as Section 7.2 shows, the direction of the ambiguity-versus-squint dependence in our model is the opposite of that literature anchor, and the contrast is a result of this paper rather than a contradiction to be smoothed over. The standard ambiguity and resolution relations otherwise follow the textbook treatment (Cumming and Wong, 2005).

The dwell time itself is treated in the imaging literature as a deterministic function of mode and geometry — Kim et al.'s synthetic-aperture time chain is exactly that (Kim et al., 2021) — but it stops at the imaging problem boundary: in the imaging references this paper draws on, the scheduling layer above it is not part of their treatment.

2.2. Agile Earth-observation satellite scheduling

The agile-scheduling literature is broad and mature: hybrid genetic-simulated-annealing planners for moving-target observation (Qin et al., 2024), deep-reinforcement-learning planners for large mission-planning instances (Long et al., 2024; Chen et al., 2024; Zhou et al., 2025), adaptive large-neighborhood search with learned guidance or with neighbourhood repair (Du et al., 2025; He et al., 2018), constraint-programming formulations for super-agile platforms (Caleiras et al., 2026), memetic and multi-chromosome cooperative planners (Wang et al., 2024a,b), and profit models in which the observation value depends on the interval between revisits (Li et al., 2024; Peng et al., 2019). Across the agile-scheduling formulations surveyed here, the dominant structural assumption is that the imaging duration of a task is an exogenous constant of the task, known before the scheduler runs — the SAR-specific line below is the nearest neighbour, and even there the geometry is fixed upstream of the planner. The scheduler then packs fixed-length intervals into time lines, and whatever the objective — profit, coverage, latency — the duration never feeds back.

For an agile SAR platform in sliding-spotlight operation that assumption is false in a specific, quantifiable way: the dwell is produced by the mode physics (the tan-exact closure chain of Section 3.2), it varies per task with the assigned mode and the squint of the selected start, and it enters the inter-task transfer constraint that decides what else the pass can carry. This paper makes the dwell endogenous, carries the required resolution as a per-task constraint on that dwell, and measures what tightening the requirement costs in coverage.

2.3. Quality-aware scheduling

Image quality reaches the planning literature in three roles, and the difference between them and this paper is worth stating plainly. Image-quality research on the SAR side carries a substantial assessment literature — fidelity metrics, quality checkers, and radiometric corrections applied to formed images (Cumming and Wong, 2005; Mittermayer et al., 2003; Prats et al., 2010). Imaging-mode design starts from a resolution requirement and trades it against swath (Song

et al., 2025), and mission-level SAR planning computes the required synthetic-aperture time and steering span from the required resolution and the incidence angle (Kim and Chang, 2020). Multi-objective Earth-observation and SAR mission studies optimise more than one criterion explicitly alongside coverage or priority (Chang et al., 2021; Kim and Seong, 2019; Kim and Chang, 2020); that line of work exists, it is not the line this paper takes, and nothing here claims its absence. What changes here is the *placement* of quality. It leaves the objective set: the per-mode index of Section 3.3 states a platform capability and supplies the numerator of a reported fulfillment ratio, and it carries no selection pressure on the search.

2.4. Positioning

The three lines meet at a seam rather than at a gap, and the seam is narrow. The imaging line supplies the resolution-to-aperture chain and stops at the imaging problem boundary: a required resolution and incidence angle fix the synthetic-aperture time and steering span of a designed observation (Kim and Chang, 2020), and the sliding-spotlight processors deliver such a design at a fixed rotation centre (Mittermayer et al., 2003; Prats et al., 2010). The scheduling line supplies the packing machinery for tasks whose duration is an attribute fixed before the schedule is built (Chen et al., 2024; Qin et al., 2024). The quality-aware line supplies assessment calibers and mission-level weightings (Cumming and Wong, 2005; Kim and Seong, 2019). This paper puts the required resolution into the scheduling constraint set: the mode parameters follow the requirement where the mode has them, the resolution delivered at the selected start equals the requirement, and the coverage cost of tightening the requirement is measured on a frozen grid under a fixed mode assignment (Section 3 and Section 6). Sections 3–6 build that problem on a registered, preregistered footing — the demand grid, the shape criteria and the caliber labels are fixed before this world's data were read — and report what the tightening costs.

A recent SAR-specific line is the closest neighbour and is worth separating explicitly. Liu et al. (Liu et al., 2025) and Xin et al. (Xin et al., 2024) plan

Table 1: Positioning on the three axes the paper treats as one chain: what fixes the imaging duration, where image quality enters, and who decides the imaging mode.

Work	Duration	Quality	Mode
Agile EO scheduling (Long et al., 2024; Qin et al., 2024; Chen et al., 2024)	exogenous constant	outside the planner	fixed by design
SAR regional scheduling (Liu et al., 2025; Xin et al., 2024)	exogenous input	not a scheduling readout	fixed by design
Multistrip planning (Shen et al., 2024)	derived from strip design	energy constraint	designed per strip
Window screening (Li et al., 2026)	generation-side pruning	screening filter	fixed per request
SAR imaging design (Kim et al., 2021; Song et al., 2025)	mode + geometry at design time	design criterion	designed, not scheduled
This paper	endogenous per task and start	reported reading, never optimised	frozen per task by its requirement

SAR regional observation with weighted multi-objective and genetic-algorithm planners; Shen et al. (Shen et al., 2024) plan multistrip stitching missions under an onboard energy budget; Li et al. (Li et al., 2026) generate visible windows and screen them for quality before any planning runs. Across these four, the imaging geometry is fixed upstream of the scheduler — by the request, the strip design, or the generation-side screen — and none carries a per-task dwell that responds to a per-task resolution requirement inside the scheduling loop. Table 1 positions the neighbouring work on the three axes this paper treats as one chain.

3. Problem Formulation and Scheduling Model

This section formalises the observation geometry, the three-mode imaging library, the dwell-duration closures, the per-mode image-quality model, and the constraints and objectives of the scheduling problem.

3.1. Observation geometry and imaging modes

We adopt the X-band reference platform of Kim et al. (Kim et al., 2021): altitude $h = 570$ km, inclination $45°$, wavelength $\lambda = 0.031$ m, azimuth beam width $\theta_a = 0.36°$, and a square scene of 5 km side length. The derived geometry is $D = \lambda/\theta_a = 4.9338$ m (antenna azimuth length), aperture-centre slant range $R_0 = 775.4$ km, and inertial scene velocity $v_s = 7578.05$ m/s. This is the platform inertial velocity, registered as the scene-traversal velocity of this scenario world rather than the beam-footprint ground velocity; it enters every dwell closure and the Doppler bandwidth as the single frozen value, so cross-mode and cross-squint dwell ratios are invariant to the choice, while absolute dwell durations and every reading that mixes dwells with velocity-independent constants (the coverage readings) or evaluates the ambiguity ratio at a fixed PRF carries the convention. Each task is a scene; the scheduled quantity is *when* to image it, not *whether* to reconfigure the payload.

The three imaging modes — stripmap (STR), sliding spotlight (SLS), and spotlight (SPT) — are treated as a single-parameter continuum over the position of a fixed virtual rotation centre (Kim et al., 2021; Zhang et al., 2018; Song et al., 2025): writing k for the rotation-centre parameter and $A = 1 - k$ for the sliding factor, $k \to 0$ places the rotation centre at infinity (stripmap, parallel antenna axis), $0 < k < 1$ places it below the surface beyond the scene (sliding spotlight), and $k \to 1$ places it at the target (spotlight). The registered operating point of this study is $k = 0.797317$, i.e. a rotation-centre-to-scene-centre distance $r_0 = 197\,119$ m ≈ 197 km. We stress the symbol discipline that the two distances are dimensionally identical but semantically distinct: $r_0 \approx 197$ km is the rotation-centre-to-scene-centre distance, whereas $R_0 \approx 775.4$ km is the

aperture-centre slant range; their ratio is $r_0/R_0 = 0.254207$, and the two are never interchangeable (Fig. 1).

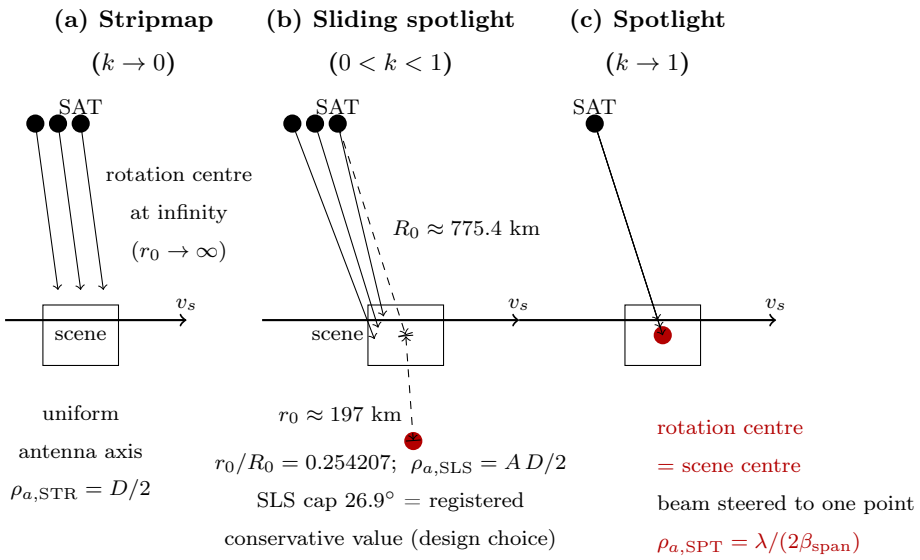
The sliding factor: under the rotation-centre semantics adopted here, $A = r_0/(R_0+r_0)$ (Zhang et al., 2018); the form $A = 1-R_0k_\theta/v_g$, with k_θ the azimuth steering rate and v_g the ground speed (Song Eq. (1)) (Song et al., 2025), is a restatement within the same rotation-centre model — for the steering rate that places the rotation centre at r_0 — not an independent literature identity. The per-mode azimuth resolutions at zero squint are $\rho_{a,\mathrm{STR}} = D/2 = 2.4669$ m, $\rho_{a,\mathrm{SLS}} = A\,D/2 = 0.5000$ m, and $\rho_{a,\mathrm{SPT}} = \lambda/(2\beta_{\mathrm{span}}) = 0.3000$ m with $\beta_{\mathrm{span}} = 2.9603°$ the spotlight sweep span (library design constant).

Under the rotation-centre semantics the spectral compression factor is a constant at every squint. Presenting the Doppler rates in the aperture-range form,

$$K_a = \frac{2v_s^2\cos^3\bar{\psi}}{\lambda R_0}, \qquad K_c = \frac{2v_s^2\cos^3\bar{\psi}}{\lambda\,(R_0+r_0)}, \qquad a_{\mathrm{res}} = 1-\frac{K_c}{K_a} \equiv A,$$

the $\cos^3$ factors cancel at beam-centre crossing, so a_{res} does not degrade with squint; the sliding-spotlight design azimuth resolution is the constant $A\,D_a/2$ (Zhang et al., 2018; Song et al., 2025).

The registered squint domains are separate registered fields, listed apart from the literature task envelope of Kim et al. ($\pm12°$, their Table 2 (Kim et al., 2021)): STR and SPT carry a design domain of $\pm45°$, and SLS carries a registered conservative envelope of $\pm26.9°$. The SLS cap is a *registered conservative value (design choice)* — antenna steering margin plus frozen-data continuity — and not a physical boundary, carried as a design choice under those two considerations with no analytic derivation claimed: under the rotation-centre semantics $a_{\mathrm{res}} \equiv A > 0$ at every squint, so high-squint reachability is limited by the registered cap rather than by mode physics. Two extrapolation disclosures attach to the $45°$ design domain: the azimuth-resolution projection correction (resolution-ellipse rotation) is outside the closed form, and the $45°$ dwell readings of STR/SPT are extrapolated, because the planar tan-difference closure omits Earth curvature against the full-curvature R_0 (Section 7.3).



Design squint envelopes (separate registered fields): STR/SPT ±45° (design domain; 45° dwell readings are extrapolated); SLS ±26.9° (registered conservative value, design choice); Kim ±12° = literature task envelope Kim et al. (2021). Under the rotation-centre semantics $a_{\mathrm{res}} \equiv A > 0$ at every squint: high-squint reachability is limited by the registered envelope, not by mode physics.

Figure 1: Three modes as a single-parameter continuum over the virtual rotation centre (azimuth squint plane, schematic). (a) Stripmap: rotation centre at infinity, uniform antenna axis. (b) Sliding spotlight: rotation centre beyond the scene ($k = 0.7973$); $r_0 \approx 197$ km = rotation-centre-to-scene-centre distance, $R_0 \approx 775.4$ km = aperture-centre slant range, $r_0/R_0 = 0.254207$; the SLS cap 26.9° is a registered conservative value (design choice), not a physical boundary. (c) Spotlight: rotation centre at the scene centre. Design squint domains listed as separate registered fields: STR/SPT ±45° (design domain; 45° dwell readings are extrapolated), SLS ±26.9° (registered conservative value, design choice), and the ±12° literature task domain of Kim et al. (Kim et al., 2021).

3.2. Dwell duration and visibility windows

The dwell of a task is the synthetic-aperture time (SAT) at scene level: the time the beam needs to sweep the full square scene at the task's squint, where W_a denotes the scene (square) size — 5 km on the reference platform

— entering as a scene-length traversal term. The production closures are the tan-exact Kim Eq. (22)/(24) chain (Kim et al., 2021). For STR the scene dwell is the exact tan difference of the beam-edge crossing angles plus the scene traversal W_a/v_s; for SLS it is the symmetric construction of Eq. (22)/(24) with the task-level sliding factor $A_{\text{task}} = 2\rho_{\text{req}} \cos\psi_{\text{sel}}/D$ in the denominator; for SPT it is the exact Eq. (10)–(11) tan-difference with the task-level sweep span $\beta_{\text{task}} = \lambda \sec\psi_{\text{sel}}/(2\rho_{\text{req}})$. The SLS and SPT closures are evaluated at these task-level parameters derived from the requirement (constraint C5, Section 3.4), so the delivered resolution equals the required resolution by construction and a tighter requirement spends more dwell; STR keeps its library-floor values, and the mode assignment itself is frozen under the demand scan (grandfathered, Section 3.5). A planar closed form with the $\sec^2\bar{\psi}$ squint correction, $T_{\text{SLS}} = (R_0\theta_a + W_a)/(A\,v_s)\cdot\sec^2\bar{\psi}$ (scene size inside the correction), survives as the test-time oracle: at the Kim reference squint of 12° it deviates from the production chain by +0.4185%, and the deviation grows to +2.1% at the SLS cap and +5.3% at 45° (an extrapolated-domain reading, Section 3.1), which is what triggered the kill-test switch from the planar form to the tan-exact chain in the first place. A frozen window-tail pad of 1% of the dwell is carried as a conservative margin on the production path; it is not claimed to bound the production closure's own squint error, which is not separately quantified here. Two provenance notes complete the closure. First, the Kim Eq. (22)/(24) chain is a zero-Doppler-clamped (broadside) attitude construction; applying it at nonzero squint is a registered approximation, and the growth of the planar oracle's deviation with squint, quantified above, bounds the oracle's linearisation error rather than the production chain's — it is evidence that the broadside-derived closure carries a modelling gap rather than numerical noise. Second, the resulting dwell feeds the inter-task transfer requirement (constraint C2, defined in Section 3.4) that determines pass capacity, which is why the deviation pattern and the frozen window-tail pad are carried with the constraint semantics (Section 7.3) rather than with the numerics.

The zero-squint anchors close an identity: $\theta_a/A = \lambda/(2\rho_a)$ holds at the

library design point ($0.006283185307/0.202683397006 = 0.031000000000$ rad $= \lambda/(2 \times 0.5\,\mathrm{m})$) with a full-precision residual of 3.5×10^{-18}. The surviving guard of this identity is a dedicated regression test in the duration-model suite.

Each task carries one or more visibility windows produced by ground-track pass enumeration over the scenario span; a window fixes the closest-approach epoch t_{ca}, the per-window slant range, and the earliest/latest feasible imaging start. The decision variable $\tau \in [0, 1]$ of a selected task is mapped onto the feasible start interval of its window (Section 4). The squint at the selected start, $\bar{\psi}$, is measured at the window's geometry.

One scheduling-level geometric quantity needs a definition here because it enters the inter-task constraint: the maneuver angle between consecutive tasks i, j, $d_\psi = |\psi_i - \psi_j|$, is the difference of their start squints. This is a planar-caliber *lower bound* on the true attitude excursion: it omits the sweep *during imaging* (the antenna keeps steering at 0.4233–0.6686 °/s, i.e. 1.7–2.7° over a 4 s dwell) and the start-squint difference understates the off-vertical attitude compression by 15–25%. The full disclosure of the two-sided understatement and the margin-pad argument is given in Section 7.3.

3.3. Image quality model

The per-task quality index is an inverse-pixel-area form with a range-normalised (non-normalised, m^{-2} caliber) readout,

$$f_{q,i} = \frac{1}{\rho_{\mathrm{rg}}\, \rho_a(m_i)}\, g_T(m_i) \left(\frac{R_0}{R}\right)^3,$$

where ρ_{rg} is the ground-range resolution, $\rho_a(m_i)$ the zero-squint azimuth resolution of the assigned mode, and g_T the transmit-path antenna gain term. The $(R_0/R)^3$ factor carries the range dependence of the radiometric term: two-way spreading falls as R^{-4} while the resolution cell grows linearly with range, so the noise-equivalent backscatter referenced to that cell scales as R^3, and the factor is therefore a *relative* normalisation carried without calibration; at $R = R_0$ the identity chain collapses to $f_q = (g_T/\rho_a) \cdot (1/\rho_{\mathrm{rg}})$ (Song et al., 2025). A scope note on the name: the index is dominated by its azimuth-resolution term, and no

calibrated radiometric quantity — noise-equivalent sigma zero, signal-to-noise ratio, or radiometric resolution — enters the index: the only radiometric-shaped factor is the uncalibrated relative range dependence above, and it is evaluated at each window's own slant range, so the index of a task depends on which of its windows is scheduled (Section 7.3); the ambiguity ratio enters separately as an admission constraint, not as a traded objective. "Image quality" in this paper therefore denotes spatial-resolution quality under a gain-shaped multiplier.

The index has two roles in this paper, and neither of them is an optimization target. It states a platform capability, and it supplies the numerator of a reported reading. As a capability statement, at the zero-squint library caliber the per-mode constants are 0.8107 (STR), 19.7352 (SLS) and 54.8322 (SPT) m^{-2}: magnitude ratios of 24.3× (SLS over STR), 67.6× (SPT over STR) and 2.8× (SPT over SLS). Under the registered fixture two of these constants also admit a direct recomputation form: the stripmap and sliding-spotlight dwell gains carry the exact product identity $g_T(m)\,\rho_a(m) = D/2$, so with the held $\rho_{\mathrm{rg}} = 0.5$ m they are equivalently D/ρ_a^2; the spotlight gain is the tan-exact point-dwell ratio rather than the tier ratio, so its constant sits a fraction of a percent above that form. Because the squint factor cancels inside the gain term (exactly for STR and SLS, and to within 0.07% for SPT across the full envelope), these constants — not squint-dependent readings — are what separates the modes on this axis, and they are properties of the mode library rather than results of a schedule.

The reported reading is a fulfillment ratio. For a served task it is the f_q at the realised window and the selected start's squint, divided by the best f_q the task could reach over all of its windows at zero squint, so the ratio is at most one. It is computed on the library caliber, averaged over all tie-set members rather than read off the anchor solution, and reported as a reading with no optimization role. Its label travels with every number it states: *[library caliber f_q | per-task f_q at the realised window and squint over the zero-squint optimum over all windows | mean over all tie-set members | reported reading, not an optimization target].*

Three caliber statements are frozen with this definition. First, ρ_{rg} is held

at the design-anchor constant of 0.5 m (registered at 45° incidence, scene-centre semantics); it is not recomputed per task, and the magnitude narrative carries a limitation (Section 7.3). Second, g_T is *self-derived* and *internally verified*: no external literature source carries the noise-equivalent sigma zero (NESZ) equation set for this mode family, so the verification route is the internal identity chain above (and its regression guard), and the absence of external cross-validation is disclosed. Third, the squint-dependent achieved resolution $\rho_{a,\mathrm{ach}}(m,\bar{\psi}) = \rho_a(m,0)/\cos\bar{\psi}$ is registered for reporting; the router consumes the zero-squint constant, and the only bandwidth consumer of $\rho_{a,\mathrm{ach}}$ is the Doppler bandwidth $B_D = v_s\cos\bar{\psi}/\rho_a(m,0)$ — there is no consumer of the achieved resolution in mode routing. The projection relation behind the anchor is $\rho_{\mathrm{rg}} = c/(2\,B_{\mathrm{rg}}\sin\eta_i)$ with the range bandwidth B_{rg} frozen at the design incidence; the planar evaluation layer holds the incidence at that design value, which is why ρ_{rg} enters the objective as a constant, with the incidence-band variation disclosed separately in Section 7.3.

The azimuth ambiguity-to-signal ratio (AASR) is defined with the ambiguity sum *excluding* the *zeroth* (principal-image) term,

$$\mathrm{AASR} = \frac{\sum_{m\neq 0}\int_{-B_D/2+f_{ac}}^{B_D/2+f_{ac}} \left|G\big(f_a + m\cdot\mathrm{PRF}\big)\right|^2 df_a}{\int_{-B_D/2+f_{ac}}^{B_D/2+f_{ac}} \left|G(f_a)\right|^2 df_a},$$

following the standard convention (Cumming and Wong, 2005): keeping $m = 0$ would place the principal image in its own numerator and force AASR ≥ 1 identically. The frequency-axis rescaling by $(1-K_c/K_a) \approx (1-k_\theta R_0/v_g)$, a flat-ground bridge between the K_c and v_g forms per the frozen formula registration, under rotation-centre steering is what couples the steering geometry into the ambiguity integral (Song et al., 2025). The displayed integral is Song's general form, with the absolute-centre shift f_{ac} carried in the integration limits; the implementation integrates the same sums about the DC-relative coordinate, where that shift cancels and the absolute centre drops out (Section 7.3). A footnote on the spotlight special case: Song et al.'s spotlight AASR (Song et al., 2025) is the degenerate integral of the same form in which neither numerator nor

denominator contains f_a after the rescaling, which is why the SPT ambiguity in our model is squint-invariant. Integration caliber: the ambiguity sum runs over the azimuth-frequency axis alone — the two-way pattern enters as a function of azimuth frequency, with no cross-range (elevation-plane) pattern factor — so far-ambiguous-pulse attenuation from the elevation plane is not modeled; the principal effect of that simplification falls on the absolute ambiguity level, and it is registered with the model caveats (Section 7.3).

Admission uses a *discrete* worst-case readout, *not continuous* evaluation: the AASR of a task is evaluated at 11 equally spaced *in-window* samples and the maximum must not exceed the registered -13 dB admission threshold at generation time. Feasibility statements in this paper inherit that caliber — a selection is feasible with respect to the generator's 11-point admission rule, and the evaluator performs no AASR recheck of its own.

3.4. Constraints and objectives

The decision vector of a scenario with N tasks is $x = (s, \tau) \in \{0,1\}^N \times [0,1]^N$: a selection bit per task and a start-time fraction per selected task. The imaging mode m_i is *not* a decision gene: the pair (required resolution, scene size) is mapped to a mode by the library table, a *system design input with product requirement semantics* — the platform is told what product a customer requires — and the mode assignment is frozen for the data of this paper (Section 3.5). What the scheduler does inherit from the requirement is the mode's parameterisation: for sliding spotlight and spotlight the design parameters are functions of ρ_{req} (constraint C5), so the requirement reaches the dwell through the mode rather than through a decision variable. Joint mode-and-time allocation is future work.

Five constraint families apply.

- **C1 (geometry, generation side).** A selected start must fall inside a visibility window of the task whose node satisfies the serviceability evidence: *every* node in the dense domain carries *at least one* in-band transit over

the span (25/25 nodes over 72 h, per-node reading taken on the registered domain construction).

- **C2 (inter-task transfer).** For consecutive tasks in start-time order, the gap must cover $(1 + 0.25)\, T_{\mathrm{dwell},i} + t_{\mathrm{man}}(d_\psi, \omega_{\mathrm{start}}, \omega_{\mathrm{end}}) + t_{\mathrm{settle}}$: the 25% pad multiplies the *dwell* (Kim's own margin semantics, stated in four places in Kim et al. 2021), never the maneuver, and the settle term is an unpadded measured constant ($t_{\mathrm{settle}} = 2.0$ s). Production evaluates t_{man} as a registered hybrid: the analytic model stands alone only where it reaches at least half of the measured calibration anchor (1.84 s, Kim scenario 1); below that trigger the per-transition maneuver bound falls back to the worst measured required maneuver time (4.07 s, pure maneuver, Kim scenario 2) — a conservative substitution that raises a visible warning in the implementation. The conflict predicate uses the bare dwell: pads are scheduling margin, not physical occupancy.

- **C3 (energy/duty).** Per-task duty is normalised against the mode-library envelope reference $T_{\mathrm{env,max}} = 10.5830$ s (the largest feasible scene dwell at the envelope-edge squint, attained by SPT) with mode weights STR/SLS/SPT = 1.0/1.15/1.3; the normalisation denominator is the largest weight times the envelope reference, $1.3\, T_{\mathrm{env,max}}$ (the C4 reference below omits this weight factor).

- **C4 (solid-state memory).** Per-echo bytes follow PRF$\cdot T \cdot N_r \cdot b_q \cdot 2/8$ with $N_r = 10\,000$ range samples and $b_q = 3$ bits per component (TerraSAR-X-style block-adaptive quantisation), normalised against the same per-window reference as C3 (each measured window's own slant range, the caliber registered for this scenario world) under the single production PRF of 3630 Hz.

- **C5 (required resolution, evaluation side).** Every selected task must be imaged at its required azimuth resolution ρ_{req}. For sliding spotlight and spotlight the mode parameters are derived from the requirement at

the selected start's squint ψ_{sel} — $A_{\text{task}} = 2\rho_{\text{req}} \cos\psi_{\text{sel}}/D$ and $\beta_{\text{task}} = \lambda \sec\psi_{\text{sel}}/(2\rho_{\text{req}})$ — so the delivered resolution equals the requirement by construction; the identity is re-verified at every dwell evaluation against a 10^{-12} tolerance, and a violation raises instead of being scored. The claim is pointwise at that start: inside the imaging interval the delivered resolution follows $\rho_{\text{req}} \cos\psi_{\text{sel}}/\cos\psi(t)$, so this paper does not claim a constant delivered resolution across a whole imaging pass. The stripmap mode (STR) delivers the library floor of 2.4669 m: it has no task-level parameter, so a requirement finer than the floor has no admissible mode for that task, and a tighter requirement there is a step at the floor rather than a continuous degradation.

Two structural facts about C3/C4 are disclosed as design semantics rather than discovered: both budgets are *structurally never binding* inside the squint envelope (their normalised values cannot exceed 1.0 there, so the binding channels are C2 and interval conflicts), and the task priorities that drive the objective are drawn uniformly at random as randint$(1, 10)$, i.e. they carry *no physical* provenance — they are demand-side bookkeeping, not measured value. Across the five constraint families, the set that actually binds in this scenario world is exactly two: the C2 transfer chain and the stripmap floor. C1 filters at generation time, C3/C4 are structurally slack inside the envelope, and C5 is a raise-guarded identity rather than a scored constraint; the inventory above is the registered model, not a claim that five mechanisms bind. Across the five constraint families, the set that actually binds in this scenario world is exactly two: the C2 transfer chain and the stripmap floor. C1 filters at generation time, C3/C4 are structurally slack inside the envelope, and C5 is a raise-guarded identity rather than a scored constraint; the inventory above is the registered model, not a claim that five mechanisms bind.

The problem maximises the served priority fraction $f_1 = \sum_{i \in S} p_i / \sum_i p_i$ over the selections S that satisfy C1–C5. Coverage is the only objective. The quality index of Section 3.3 carries no selection pressure: the search optimises

one criterion rather than two, a front is a tie-set on coverage, and the index appears only as the fulfillment reading. The search method is described next.

A reader who meets this model through earlier versions of this line will expect a quality aggregate to be traded against served priority, so the change is stated explicitly. This is not a re-parameterisation of the same problem: the search returns a tie-set instead of a front, and the quality index moves from a decision criterion to a measured property of whatever the scheduler serves. The change is registered as an amendment to the project's preregistration, which vacates the quality-headroom estimand and registers the coverage–demand curve of Section 3.5 as the main result form.

3.5. The demand-tightening axis

The experimental axis of this paper is external to the scheduler. A scalar s multiplies every task's required resolution, $\rho_{\mathrm{req},i}(s) = s\,\rho_i^{\mathrm{req}}$, with $s = 1$ reproducing the requirement as placed and $s < 1$ a tighter requirement; nine values are frozen, $s \in \{0.8, 0.9, 1.0, 1.1, 1.2, 1.3, 1.4, 1.5, 2.0\}$. Scaling touches the requirement only: the scenario population — geometry, task set, and visibility windows — is not regenerated, so every point of the grid is one scenario world seen through a different requirement.

The axis is swept under a frozen mode assignment. Which mode a task receives is not re-derived as the requirement tightens: the assignment is the one the scenario world was generated with, inherited unchanged (the grandfather decision registered in the project's frozen formula registry, formula-spec §8.17-9). The reason is comparability — re-cutting the mode mix at every grid point would confound the coverage response with a routing change — and it fixes the scope of every conclusion that follows. The results are requirement sensitivity under a fixed mode assignment, not a general statement about how coverage trades against resolution when a router is free to move tasks between modes; the experiment that would answer the general question is a different one and is not run here.

The mode assignment matters most on the stripmap family, because that

is where the floor sits. A family whose tasks are all stripmap cannot respond to a tighter requirement continuously: the delivered resolution is the antenna floor, so the family either serves at the floor or finds no admissible mode. The single-satellite family R1C meets that floor inside the frozen grid — at $s = 0.8$ the family has no feasible individual in 10 of 10 seeds — and that outcome is an infeasibility record, not a measured coverage of zero. The two statements are kept apart everywhere in this paper: an infeasible grid point is not a point on the curve, and it is never plotted, averaged, or quoted as one.

Three different mode rules are in play in this project, and a reader should not mistake one for another. (i) The admission path routes a requirement to the coarsest library tier whose zero-squint resolution meets it (the coarsest-sufficient tier routing, `route_mode`); that rule consults the requirement and the library alone, one tier per task, with no scene-size test. (ii) The registered selection policy adds the feasibility test ($A_{\text{task}} \in (0, 1)$, the mode envelopes, the spotlight footprint bound, the ambiguity gate) and a two-key rule on scene size — spotlight inside the footprint, sliding spotlight beyond it, stripmap at the coarse end — and is the policy registered for narrative and for future families (formula-spec §8.17-9). (iii) The data of this paper follow neither of those at run time: they carry the grandfathered assignment described above. Where this paper says that the mode follows the requirement, it means that the parameters of the *assigned* mode follow the requirement, not that the assignment itself changes.

Coverage is measured on this axis as a reading: for each family and each grid point, the median over ten seeds of the served priority fraction of the best-covered member, at a single search budget. Three shape criteria are frozen against the curve — monotone non-decreasing medians across the grid, a saturation arm ($s = 2.0$ at least $s = 1.5$), and a tight arm ($s = 0.8$ below $s = 1.0$) — and they are inherited unchanged from their earlier registration rather than tuned to this world's data. Section 6 reports them as measured, failures included. Coverage admits no pass/fail verdict here: every unit's front is a tie-set carrying one coverage value, so there is nothing for a verdict to decide, and the

curve is reported as a reading.

4. Solving Approach

In scheduling-theory terms the problem of Section 3 is a single-machine scheduling problem with time windows and sequence-dependent setup times: the satellite is one resource, each task carries visibility windows, and the inter-task transfer of constraint C2 charges a maneuver time that depends on the attitude endpoints of the consecutive pair — a sequence-dependent setup in the classical sense. The imaging dwell adds a second classical construct: because the dwell follows from the mode and the squint of the selected start, a task's processing time is not a constant but a function of where it is started within its window, a position- and time-dependent processing time of the kind studied in the time-dependent scheduling literature (Gawiejnowicz et al., 2009; Gawiejnowicz and Kurc, 2015; He et al., 2019). No complexity claim is made in this paper: no hardness proof or approximation result is attempted, and the evolutionary search of this section is offered as a measurement instrument on a registered scenario world, not as a contribution to scheduling theory. Bounding methods for the fixed-dwell agile-observation problem exist (Habet et al., 2010); how they transfer when the processing time is endogenous is an open question this paper does not address.

4.1. Encoding and evaluation

The solver is a single-objective evolutionary search over the chromosome of Section 3.4: a selection bit and a start fraction $\tau \in [0, 1]$ per task — no mode gene, since the mode is fixed by the product requirement — so the densest real-window family (R3H, defined with the scenario families in Section 5.2) encodes its 99–100 admitted tasks (of 100 drawn) over 731–779 enumerated windows as $2N$ decision variables with $N = 99$–100; at the coarsest end the control family (R1C, 20 drawn and 20 admitted) encodes 40. Decoding projects τ onto the union of feasible window intervals of the admitted windows: a served

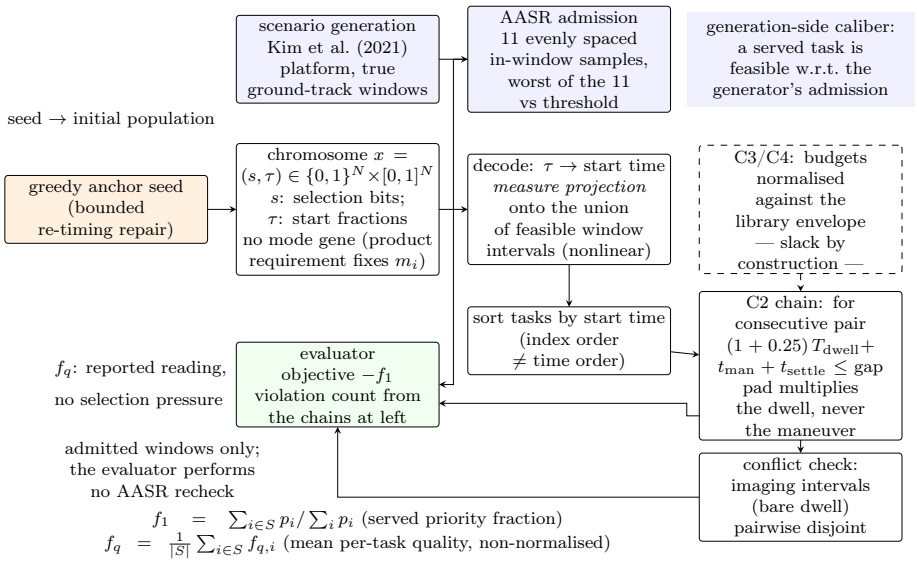


Figure 2: The scheduler's encoding-evaluation loop. Generation side (blue): scenario construction and AASR admission (11 evenly spaced in-window samples, worst-of-11 against the threshold). Genome: selection bits and start fractions, no mode gene; the greedy anchor seed is injected into the initial population. Decode is a measure projection of τ onto the union of feasible window intervals (nonlinear by construction). Evaluation: tasks are sorted by start time (index order is not time order), the C2 chain charges the $(1 + 0.25)$ pad on the dwell — never on the maneuver — plus the maneuver and settle terms, conflicts use the bare dwell, and C3/C4 (dashed) are slack by construction inside the envelope. The evaluator itself performs no AASR recheck: feasibility is relative to the generator's admission caliber.

task is feasible with respect to the generator's discrete 11-point admission rule (Section 5.1), and the evaluator performs no AASR recheck of its own; feasibility statements in the results are relative to that generation-side caliber, not to an independent physical guarantee (Fig. 2).

4.2. Anchor-seeded initialization

The initial population contains the greedy anchor: a bounded single-pass re-timing repair of the all-selected schedule. The seed is what makes the f_1 normalisation meaningful — every front carries a solution anchored at the served-

priority reference — and its injection into the population is checked by a dedicated cross-check that once exposed spurious violations from time-inverted index pairs, which is also why the evaluation loop sorts by start time before charging the C2 chain.

4.3. Search operators and selection

Variation applies simulated binary crossover ($\eta = 15$, per-variable probability 0.9) and polynomial mutation ($\eta = 20$) uniformly over the real-valued chromosome — the selection bits are ordinary real genes decoded by a 0.5 threshold, so no separate binary operator acts on them. Because the problem carries one criterion, the non-dominated sorting of the standard machinery ranks by that single value alone (Deb et al., 2002), and the survivor set of a unit is a tie-set on coverage rather than a trade-off front: the search returns the best-covered selections it reaches and the tie-set records their common coverage (Section 3.3). The search runs under a fixed random seed per unit, single-threaded, with manifest task-list order everywhere (no set or dictionary iteration order may influence a run), so every unit is bit-reproducible.

4.4. The measure normalisation of τ and its implicit bias

One parameterisation choice is disclosed as intentional. A start fraction τ indexes position within the *feasible measure* of a task's window union, not within wall-clock time: decoding is the measure projection of Fig. 2. Because the projection is measure-normalised, a uniform grid on τ samples wider windows more finely — the parameterisation carries an implicit weighting toward wider windows. This bias is registered as a deliberate design decision: it is not part of the main specification or its preregistration, and its registration is carried in the project's dated design-amendment record (review note of architecture decision AD-3). No selection mechanism in the loop undoes the weighting; it is part of the search design, and every reading in Section 6 inherits it.

4.5. The search face and the construction face

The response this paper reports has two faces, a search face and a construction face, and the separation between them is measured rather than assumed. Over the 260 units that produced an artifact (nine demand points × three families × ten seeds, less the ten exit-11 units of Section 6.2, where the search exits with code 11 when no feasible individual exists for the unit), the greedy anchor already attains the best coverage found for the unit in this study in 258; the search exceeds its own anchor in exactly two units — the control family R1C at the as-placed point, seeds 100 and 105 — by +0.0427 and +0.0463 of the served priority fraction. That is a statement about where the coverage response comes from, not a statement about solver quality. The chromosome can do only two things: move a start inside the feasible measure of its window union, and drop tasks from the selection. Which tasks remain serviceable at a given requirement is therefore decided before the search runs, by the mode assignment and by the parameter adaptation that carries the requirement into the dwell; the search's job is to reach the level that admission sets. Within this world the anchor is what reaches it, and the two units where the anchor does not are recorded rather than smoothed. The consequence for reading Section 6 is direct: the coverage differences between demand points are a property of the construction, and no statement in this paper attributes them to the optimiser.

5. Experimental Design and Setup

This section describes the scenario world, the family and caliber system, and the preregistered test design.

5.1. Scenario construction

The scenario world is built from *real ground-track transit enumeration*, not from an arbitrary orbit phase point: visibility windows come from enumerating true ground-track passes over the scenario span, with a 15 s scan step, gap merging at 300 s (a registered band parameter, invariant across the registered

150/600/1200 s sensitivity band, not a physical lower bound), and 33-point edge refinement. The registry's own ledger records why: under the earlier phase-point pass rule only 0 of 8 clusters were serviceable after the incidence fix, against 6 of 8 under true enumeration. The demand mix itself — resolution requirements and scene sizes — is a registered synthetic template parameter, not a measured campaign profile; the priorities that drive the objective are uniform random integers with no physical provenance (Section 3.4), so the mode composition of the served set reflects the template's requirement distribution. The template draws each task's resolution requirement from two frozen tiers — a coarse tier spanning $[2.5, 3.0]$ m and a fine tier spanning $[0.3, 0.6]$ m — so the stripmap floor of 2.4669 m (Section 3.3) sits inside the coarse tier, and the tight end of the demand grid scales requirements below what the floor's mode can serve.

Orbit propagation is two-body only — no J_2 or atmospheric terms — so the registered repeat figure is the *simulated* ground-track repeat of 14.97 orbits $\approx$ 23.93 h (one sidereal day over the two-body period); the figure that includes J_2 nodal drift describes the physical orbit and is deliberately not the one this simulator produces. The surface normal used in incidence computations is an *approximate* normal: its maximum deviation from the true ellipsoidal normal is 0.125° (at $\approx$30° latitude; 0.109° at the dense-domain centre), with no effect on band-boundary judgements at the registered margins. The dense domain is a 1000 km square centred at 20°N/0°E, a 5×5 node grid at 250 km spacing with per-cell target jitter, a 3-day span in coordinated universal time (UTC). The serviceability criterion is per-node: every node carries at least one in-band transit within the span, measured 25/25; the literal best-incidence reading of the dense-domain constraint fails at the centre node (near-nadir passes exist there but yield no in-band window), which is why the operative criterion is the per-node one. The 25/25 reading is a construction-time measurement on the full-span domain; the manifest ledger stores aggregate admission counts, not per-node columns.

The AASR admission criterion is evaluated per family, and the two families differ by design — a discrepancy we disclose rather than harmonise post hoc.

Synthetic M-family windows are admitted by a window-edge single-point readout (the AASR at the window edge squint only), whereas the real-window R families sample the AASR at 11 evenly spaced in-window instants ($\mathrm{lo}+(\mathrm{hi}-\mathrm{lo})\,j/10$, $j = 0, \ldots, 10$) and admit the window only if the worst of the 11 samples meets the threshold. A sample outside the physical sliding domain contributes an exclusion, so it can never mask an unserviceable window. The window-edge single-point rule *underestimates* the in-window worst case by about 4 dB, so the two admission rules are stated per family wherever admission is discussed; the R-family rule is the discrete in-window worst-case caliber that all feasibility statements in this paper inherit (Section 3.3). This sampling is *discrete*, *not continuous* evaluation: a window admitted under it is feasible with respect to the 11-point rule, and the evaluator performs no AASR recheck of its own. Two further conservative narrowings bound the admissible window set and are carried with the same caliber: a pass is not admitted when the scene along-track footprint exceeds the beam-limited bound $R_0\,\theta_a$ at the closest-approach slant range of that pass (binding for SPT), and the envelope mask admits starts only inside a 1% inward guard of the mode envelope, within a $\pm$120 s span around the pass closest approach, with each admitted mask run inset by one scan step ($\approx$1 s) per end. Across the 30 released R-family manifests the generation ledger records zero tasks removed by the AASR threshold (removals: aasr 0, footprint 0, below-finest 0; no-window 15), so the threshold is part of the admission caliber without ever binding in this scenario world. Across the 30 released R-family manifests the generation ledger records zero tasks removed by the AASR threshold (removals: aasr 0, footprint 0, below-finest 0; no-window 15), so the threshold is part of the admission caliber without ever binding in this scenario world.

5.2. Families and calibers

Three real-window families form the experiment matrix, and their sizes are stated in the two numbers the registered density ladder fixes, the tasks drawn and the tasks admitted (serviceable): R1C (20 tasks drawn, 20 admitted, 145–

162 windows) is the single-satellite control; R2F (50 drawn, 46–50 admitted, 337–391 windows) and R3H (100 drawn, 99–100 admitted, 731–779 windows) are the dense-cluster treatments. Counts are stated as intervals over the ten seeds of each family; a single-seed value is not a family constant. Task counts follow the registered density ladder $N = 20/50/100$; R4H ($N = 500$) was retired at registration because the single-transit STR pass-count scaling exceeds the registered span at $N \geq 150$ (the retirement rationale is registered; no capacity number is asserted anywhere in this paper, since the serial bound has not been recomputed under the present real-window scenario world).

The mode composition of each family is inherited from the grandfathered assignment of Section 3.5 rather than re-derived at run time: every R1C task is stripmap, R2F routes its tasks between sliding spotlight and spotlight and carries no stripmap task, and R3H mixes all three modes. That composition sets the scale of the quality index a family can reach (Section 3.3), but it is a property of the demand template that generated the scenario rather than of any schedule, and no reading in this paper compares index levels across families. The control family's served fraction saturates at 1.0 from the as-placed point upwards; the saturation is anchor-construction-carried (Section 6.5) and is read descriptively, never as a performance result.

The artifacts are the project's run-lineage node L8 (caliber v5), and they are not a refresh of the previous world but an explicit re-judgement. A registered decision made the sliding-spotlight and spotlight parameters follow the requirement per task (constraint C5), so the delivered resolution at the selected start equals the requirement by construction; that changes what is admissible at the data level rather than merely how a reading is labelled, and the earlier artifacts therefore no longer describe the same model. L8 re-runs the sweep over the same frozen scenario population — the task set, the visibility windows and the mode assignment are carried over unchanged, so the demand axis is the only thing that moves — at a single registered budget of 40/60 population/generations. The project's run-lineage register records the L8 node with its parent, its frozen head and its adjudication, and every reading in Section 6 is recomputed from

those artifacts by the manuscript-number checker before the section is read as a result.

The R-family f_q is *not normalised*: the per-window slant-range geometry expands the units scene-dependently, so index levels are incomparable across families and are never pooled into a single span or compared across families. The synthetic M families remain available as a controlled reference; their one still-valid artifact is annotated, wherever cited, as a version-2-caliber controlled reference whose output provenance is unregistered, and no M-family number is mixed with R-family numbers in this paper.

5.3. The demand scan and its frozen criteria

The scan. The experiment is an external scan rather than a hypothesis test about a control: a scalar s multiplies every task's required resolution (Section 3.5), nine values of s are frozen, $s = 1.0$ reproduces the requirement as placed (the as-placed point), and each grid point is read as the median over ten seeds — ten independent scenario draws, each solved once, with the same seed driving both the scenario draw and the search, so scenario-sampling noise and search noise are not separated by the design — of the served priority fraction of the best-covered member, at the registered 40/60 budget. The scenario population is not regenerated between points, so within a seed the nine readings are nine views of one world through nine requirements. The search-noise half of that confound is bounded by the construction face: the anchor already attains the unit maximum in 258 of the 260 sweep units, so the coverage response is carried by the construction rather than by solver stochasticity (Section 4.5).

Three criteria, frozen before this world's data were read. Three shape criteria are registered against the curve and are inherited unchanged from their earlier registration rather than tuned to the data. C1 asks that the medians be monotone non-decreasing across the grid; C2 is the saturation arm, the median at $s = 2.0$ at least the median at $s = 1.5$; C3 is the tight arm, the median at $s = 0.8$ below the median at $s = 1.0$. A cell whose ten seeds all exit without

a feasible individual enters the criterion evaluation as a zero-coverage median, and it is the tight arm (C3) and the monotone arm (C1) that read that cell; the saturation arm (C2) compares the two loosest points and does not. The zero-coverage entry is a recorded outcome rather than missing data, and the reporting discipline of Section 6.2 keeps such a point off the coverage curve and never averages it into one. They are the registered content of the preregistration amendment that also vacated the quality-headroom estimand of the earlier line and registered the coverage–demand curve as the main result form (Section 3.4). Their verdicts are reported as measured, failures included, and the criteria are not re-tuned after the fact: a failed criterion is a result about this world, not a defect in the criterion to be edited away. The registration timeline is part of the record and is stated exactly: the nine-point grid and the three criteria were registered before this world's data were read; the amendment that registered the coverage–demand curve as the main result form is dated 2026-10-21, after the sweep ran on 2026-10-20, so the headline curve is a post hoc descriptive reading and the pre-registered content is the grid and the criteria themselves. The refined reading of the criteria verdicts (an infeasible cell is undefined; the tight arm reads pairwise per-seed deltas against a one-priority-unit tolerance) is likewise a post hoc refinement, registered 2026-10-23 after the 2026-10-22 external review; the frozen criteria text is not edited.

What the amendment retired. The earlier design carried an effect size, a power calculation and a power curve, all of them calibrated on the quality headroom of a front over its anchor. That estimand does not exist in this model — a unit's front is a tie-set on coverage, so there is no quality quantity to be gained — and the power figure is retired with it rather than reproduced: a power curve about an estimand this paper does not report would be a number in search of a claim. Nothing in Section 6 depends on it.

One budget, stated. Every point of the grid is read at the single registered budget of 40/60; no budget ladder is run in this world, and no frontier-richness statement in either direction is made from these artifacts.

6. Results and Analysis

All readings in this section are taken from the L8 artifacts at caliber v5 (Section 5.2) and are cross-checked against the reading ledger that stores each of them with its unit, caliber label and source path. The grid is the nine-point demand scan of Section 5.3, every point read at the registered 40/60 budget.

6.1. The coverage–demand curves

The result of the paper is the coverage–demand curve (Fig. 3): the served priority fraction of the best-covered member, median over ten seeds, as a function of the demand-tightening factor. Three families produce three different shapes, and the difference between the shapes is the result rather than a nuisance variation to be averaged away.

The 100-drawn / 99-admitted family is the one the tightening actually reaches. Its median coverage falls monotonically with the tightening: 0.4827 at $s = 0.8$, 0.6884 at $s = 0.9$, 0.9485 at the as-placed point $s = 1.0$, then 0.9766 at $s = 1.1$, 0.9812 at $s = 1.2$, 0.9882 at $s = 1.3$, 0.9882 at $s = 1.4$, 0.9900 at $s = 1.5$ and 0.9900 at $s = 2.0$ (family R3H). Tightening the requirement from the as-placed point to the tightest point that still has a feasible individual costs about half of the served priority mass; loosening it from the as-placed point to the coarsest point buys back under five points. Most of the response therefore sits on the tight side of the as-placed point, which is the side an operator would actually consider. Table 2 decomposes how much of that tight-side response is the stripmap floor rather than the dwell mechanism. The floor-exclusion upper bound — every non-stripmap task servable, each stripmap task servable only if its scaled requirement still reaches the 2.4669 m floor, priority-weighted and medianed over seeds — already caps the family at 0.748 at $s = 0.9$ and 0.501 at $s = 0.8$: the floor accounts for 20.01 of the 26.01-point drop to $s = 0.9$ and for 44.78 of the 46.58-point drop to $s = 0.8$, and the residual gap between the bound and the observed medians (6.0 and 1.8 points) is the schedule falling short of a bound that is itself floor-driven. The three families covary in mode

Table 2: Floor-exclusion upper bound and observed coverage on the nine-point grid. Upper bound: every non-stripmap task servable, each stripmap task servable only if its scaled requirement reaches the 2.4669 m floor, priority-weighted, median over the ten seeds of each family. Observed: median served-priority fraction of the best-covered member (L8 artifacts, budget 40/60). The control family at $s = 0.8$ admits no feasible individual and is kept off the curve (Section 6.2).

	R1C (all STR)		R2F (no STR)		R3H (mixed)	
s	bound	obs	bound	obs	bound	obs
0.8	0.000	—	1.000	0.976	0.501	0.483
0.9	0.569	0.540	1.000	0.974	0.748	0.688
1.0	1.000	1.000	1.000	0.976	1.000	0.948
1.1	1.000	1.000	1.000	0.976	1.000	0.977
1.2	1.000	1.000	1.000	0.996	1.000	0.981
1.3	1.000	1.000	1.000	0.996	1.000	0.988
1.4	1.000	1.000	1.000	0.992	1.000	0.988
1.5	1.000	1.000	1.000	0.992	1.000	0.990
2.0	1.000	1.000	1.000	1.000	1.000	0.990

composition as they do in task count — the control family is all stripmap, the 50-task family has no stripmap task, and the dense family is mixed — which is why the same tightening reads as a floor step, a plateau, and a floor-dominated fall (Section 5.2).

The 50-drawn / 46-admitted family is nearly flat and sits high. Its median coverage reads 0.9764 at $s = 0.8$, 0.9737 at $s = 0.9$, 0.9764 at $s = 1.0$, 0.9764 at $s = 1.1$, 0.9960 at $s = 1.2$, 0.9960 at $s = 1.3$, 0.9924 at $s = 1.4$, 0.9924 at $s = 1.5$ and 1.0000 at $s = 2.0$ (family R2F). On this family a tighter requirement is close to free in coverage: the two local declines, of 0.268 and 0.359 percentage points, are small enough that they are given their own accounting (Section 6.4) rather than read as a trend.

The control family is the single-satellite one, 20 drawn and 20 admitted, and its response is a step rather than a curve. Its median coverage reads 0.5398 at

$s = 0.9$ and 1.0000 from $s = 1.0$ all the way to $s = 2.0$ (family R1C): the family either serves the whole priority mass at the stripmap floor or serves a strict subset of it, and there is no continuous intermediate regime for it to occupy. Its tightest grid point behaves differently again and is kept apart from the curve (Section 6.2).

Read together, the three curves say what the mode assignment decides. A family whose tasks are all stripmap has a floor and therefore a step; a family that mixes sliding spotlight and spotlight absorbs a tightening in its parameters and pays almost nothing in coverage; and the family that mixes all three modes, at the largest task count, is the one on which the tightening bites. That ordering follows the construction of Section 4.5 — the modes and their parameter adaptation decide which tasks stay serviceable — and it is not a property of the search.

6.2. The as-placed point, the floor step, and the cost of a tighter requirement

Every cost in this subsection is measured against the family's own saturation point — the first grid point at which its median reaches the family's maximum — and never against another family or another world. That choice is what makes each number comparable to nothing but itself, and it is stated on every line that carries one.

Tightening is where the cost concentrates: from the as-placed point to $s = 0.9$ the median coverage drops 26.0 percentage points for R3H and 46.0 for R1C, and to the tightest feasible point 46.6 for R3H, while R2F holds within 0.27 points (0.00 at $s = 0.8$). R3H pays a coverage cost of 4.15 percentage points at the as-placed point against its own saturation at $s = 1.5$, the first point at which its median reaches 0.9900. R2F pays a coverage cost of 2.36 percentage points at the as-placed point against its own saturation at $s = 2.0$, where its median first reaches 1.0000. R1C's cost against its own saturation is 0.00 percentage points, and that zero is a statement about where the family's saturation sits — the as-placed point is its saturation point — not a claim that a tightening is free for every family. Own saturation is the grid argmax by

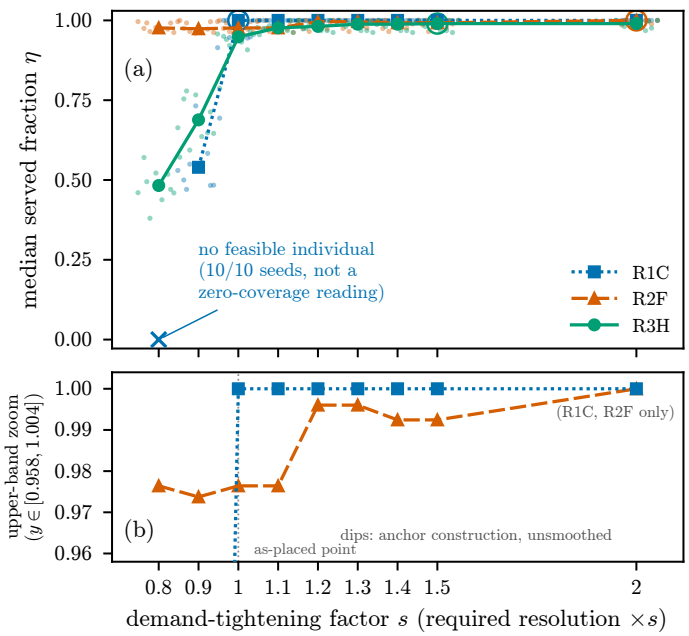


Figure 3: Coverage versus the demand-tightening factor s (L8 world, caliber v5, budget 40/60, ten seeds per point). (a) Median served priority fraction of the best-covered member; per-seed readings as dots; a hollow marker sits on a family's own maximum. The cross at $s = 0.8$ of the control family is *not* a zero-coverage reading: all ten seeds of that cell exited with no feasible individual (Section 6.2). (b) Upper-band zoom, $y \in [0.958, 1.004]$, for the two families whose local declines are at issue; the 100-task family's medians there lie below the range — its drop to $s = 0.8$ is 46.6 percentage points (Table 2). The dotted line marks the as-placed point.

construction, and its placement carries the grid: R1C saturates at the as-placed point itself, R2F's maximum sits on the grid boundary at $s = 2.0$, and R3H's plateau is flat to within one priority unit from $s = 1.4$ upwards. The premiums are read against those points, not against a grid-independent optimum.

The control family's tightest grid point is a different kind of fact and is separated from the curve on purpose. R1C at $s = 0.8$ has no feasible individual in 10 of 10 seeds: every unit of that cell exited with the no-feasible-individual code, so the family admits no admissible schedule at that requirement under the frozen mode assignment. That is not a coverage of zero, and the two must

never be interchanged. A zero would say that the search considered schedules and served none; what happened is that the requirement at $s = 0.8$ asks for a resolution finer than the stripmap floor the family is fixed to, so there is no admissible schedule to consider. The figure draws the point as a cross against the axis carrying that label, the ledger stores it as a status rather than as a value, and no average, no smoothing and no line segment passes through it. Every statement about this point in the rest of the paper keeps the same distinction.

6.3. The frozen criteria and their verdicts

Table 3 reports the three criteria of Section 5.3 under the refined reading registered on 2026-10-23: an infeasible tight cell is undefined rather than a zero-coverage median, and the tight-arm verdict reads the median of the pairwise per-seed deltas ($s = 1.0$ minus $s = 0.8$) against a tolerance of one priority unit. The frozen verdicts remain on record, and the only cell where the two readings differ is the control family's tight arm: under the frozen zero-coverage scoring convention it read "met"; under the refined reading it is undefined, because the tight cell admits no feasible individual in any seed. Why that cell is infeasible is arithmetic, not search: the control family is fixed to the stripmap mode, whose floor is 2.4669 m, and its coarse-tier requirements span $[2.5, 3.0]$ m, so at $s = 0.8$ every scaled requirement $0.8 \times [2.5, 3.0] \in [2.0, 2.4]$ m sits strictly finer than the floor — no admissible mode exists for any task, and the infeasibility is an exact consequence of the frozen bandwidth and the frozen grid. The monotone verdict is met under both readings. The 50-task family fails C1 under both; its tight arm does not fall under either reading (the two dips of Section 6.4 sit below the one-priority-unit tolerance), and both failures are reported as results about this world.

C3's failure carries information about the model, and it is the falsification of a reading rather than a null result. In the previous world this family's tight arm carried a large penalty — the median at the tightest point was 0.446 of the served priority mass — and that penalty is *withdrawn*: under the caliber-v5 model the same family reads 0.9764 at $s = 0.8$, equal to its own as-placed value

Table 3: The three shape criteria of Section 5.3, on the nine-point demand grid (L8 world, caliber v5, single registered budget 40/60, ten seeds per point), under the refined reading of the 2026-10-23 amendment: an infeasible cell is undefined, and the tight arm reads pairwise per-seed deltas against a one-priority-unit tolerance. The criteria themselves are frozen and not re-tuned; the only cell where the refined reading departs from the frozen verdicts is the control family's tight arm (frozen: met under the zero-coverage scoring convention; refined: undefined), as detailed in the text.

family (drawn / admitted)	C1 monotone	C2 saturation	C3 tight arm
R1C (20 / 20)	met	met	undefined
R2F (50 / 46)	not met	met	not met
R3H (100 / 99)	met	met	met

at the displayed precision — not below it —, so the tight arm no longer falls at all. The withdrawn value is a historical comparison and is labelled as such wherever it appears; it is not a reading of this world and it is not an estimate this paper stands behind.

What replaces it is smaller and better localized. The tightening still costs this family something, but the cost is measured at the as-placed point against the family's own saturation (2.36 percentage points, Section 6.2), not at the tight arm. The mechanism is the parameter adaptation of constraint C5: the mode parameters are derived from the requirement at the selected start, so the delivered resolution equals the requirement by construction, and a tighter requirement therefore spends dwell rather than serviceability. The family pays in imaging time; it does not pay by losing tasks, until the dwell term grows enough to collide with the transfer constraint. Saying that the cost has vanished would be wrong on both halves of that sentence: the tight-arm penalty is gone, and the as-placed cost is not.

C1's failure is the pair of local declines that Section 6.4 accounts for, and it is not removed by re-tuning. A criterion edited after its verdict is not a criterion: both failures stand as this world's answer to questions that were fixed before its data were read.

6.4. The two local declines: an anchor-construction effect

Two local declines interrupt the flat band of the 50-task family: 0.268 percentage points between $s = 0.8$ and $s = 0.9$, and 0.359 percentage points between $s = 1.3$ and $s = 1.4$. Both are attribution results, and the attribution is not the one a reader would expect.

Neither is a search miss. Over all 180 sweep units of the two treatment families the best-covered member of a unit's tie-set equals the greedy anchor's value exactly, and the search-only maximum is empty; across the whole 260-unit world the search exceeds its anchor in the two control units of Section 4.5 and nowhere else. A decline produced by a search that never misses is not a search miss. It is produced by the anchor construction itself, which is deterministic and reproducible: the anchor re-times the all-selected schedule in a single bounded pass, and at a given requirement that pass can serve one task fewer or one task more than at the neighbouring requirement.

The per-seed accounting makes the mechanism visible. Between $s = 0.8$ and $s = 0.9$ two seeds move in opposite directions — one loses 1.83 percentage points and one gains 0.42 — and the median falls because the net of those two displacements is negative. Between $s = 1.3$ and $s = 1.4$ the same shape recurs with different seeds: one loses 1.14 points and one gains 0.42. In both cases the decline is the median of ten readings following two of its members, not a property of the demand axis.

This is a different object from the search-artifact verdict of the earlier line, and the difference matters for how the reading is used. That verdict retired a reading because a larger search budget removed it: what was retired was an artifact of the search. Here no budget changes the picture, because the effect sits in the construction, which does not depend on the search at all. And nothing physical follows from either decline: two opposite single-seed displacements are not a monotonicity claim in either direction, and no statement about the demand axis is made from a movement in the third decimal place.

6.5. Demand satisfaction and retention

The curve says how much priority mass is served; a second reading says what that service amounts to. Over the nine demand points the tie set serves 477,308 task instances, distributed over 9,430 tie-set records: a served task instance is one tie-set member's selection of one task, not a distinct task — the best-covered member of each unit serves 13,845 distinct tasks in total over the grid, a factor of 34.5 below the instance count — and the count is the sum over every record of the tasks that record selects. The delivery identity that constraint C5 enforces is a per-mode rule: for sliding spotlight and spotlight the delivered resolution equals the requirement at the selected start by construction, and for stripmap the library floor is delivered at or coarser than the requirement. The identity is checked by a raise rather than by a score, so no artifact stores a per-record verdict; the observable is the constraint violation, whose maximum over every record is 0.0, and no violation is recorded relative to the evaluator for any unit of the sweep. Ten of the 270 launched units exited with no feasible individual (the control family at $s = 0.8$) and are excluded from the served count by the driver ledger rather than folded into it as zeros.

The retention reading measures the same service from the schedule's side, and it travels with a caliber that must travel with it. On the STR-clip-only caliber — retention is the fraction of the tie-set feasible measure that remains feasible after the stripmap clip (numerator: the tie-set measure still feasible after the clip, denominator: the tie-set measure before it), evaluated at the as-placed point — retention reads 51.14% for the control family, 100.00% for the 50-task family and 71.00% for the 100-task family. The earlier line's retention figures, which were computed under the retired SLS/SPT narrowing, are superseded and are not quoted here.

6.6. The fixed-dwell counterfactual

How much does the endogenous dwell actually buy? The registered counterfactual answers by scheduling against the classical assumption. For each family

and each seed at the as-placed requirement, a freeze-dwell arm plans the schedule with the dwell held at its zero-squint library value at every consumption point — the same solver, the same budget, the same scenario population as the registered arms — and the planned selection and start times are then replayed against the real, squint-dependent dwell.

The fixed-dwell plan does not survive the replay on the dense families. The transfer constraint is violated in 20–29% of consecutive pairs (median over seeds; 184 of 926 pairs for the 100-task family, 135 of 458 for the 50-task family), and after a deterministic time-order repair that drops the later task of each violating pair, the replayed schedule lands 18.25 and 30.44 percentage points (family medians) below what the endogenous dwell delivers. The control family — sparse, and all stripmap — is untouched: zero violations in nine of ten seeds and a 0.0-point median gap. That contrast is the signature of the mechanism: the fixed-dwell assumption fails exactly where the C2 chain is tight, and a classical fixed-duration scheduler would ship the plan without knowing. The counterfactual is a reanalysis of the registered artifacts rather than a new experiment; the fixed-dwell ablation of the retired v4 world (2026-10-17) is not mixed in.

6.7. Reported readings

One reading is reported beside the curve. It is not an optimization target, and it travels with the caliber label that defines the quantity it states.

The fulfillment ratio. For a served task the ratio is its f_q at the realised window and the selected start's squint, over the best f_q the task could reach over all of its windows at zero squint, so the ratio is at most one; it is averaged over all tie-set members and reported, never optimised. At the as-placed point the fulfillment ratio reads 0.6580 for R1C, 0.7764 for R2F and 0.7447 for R3H. Its label travels with every number it states: *[library caliber f_q | per-task f_q at the realised window and squint over the zero-squint optimum over all windows | mean over all tie-set members | reported reading, not an optimization target]*. What the reading says is narrow and is kept narrow: it measures how much

of the zero-squint optimum over a task's own windows survives the choice of window and start, not how good the image is by any calibrated radiometric standard (Section 7.3).

7. Discussion and Limitations

7.1. Scope and negative claims

Five scope statements are registered before anything is claimed. First, the capacity law carries a per-pass caliber: the registered serial bound is defined per pass, not per orbit, and the closure (the served fraction of the bound) is a containment reading — it says the schedule fits inside the bound, not that the bound validates the scheduler. The minimum-over-pairs transfer time that enters it is a legal lower bound on the true transfer, never an average or a worst case. Under the present real-window scenario world the serial bound has not been recomputed, so this paper reports no capacity number in any form; the law, its caliber, and its discipline are the contribution, and the recomputation is future work.

Second, the main claim is bounded by its mode assignment. The results of Section 6 are requirement sensitivity under the frozen mode assignment of Section 3.5: the curve answers what tightening the requirement costs when the mode a task receives does not change. It does not answer how coverage trades against resolution when a router is free to move tasks between modes, and that experiment is not run here. Third, the curve is read at one registered search budget (Section 5.3), so it is a reading of this world at that budget and not a statement about a converged frontier in either direction.

Fourth, the trade-off this study claims is the resolution-versus-dwell-time axis of the mode continuum. The antenna-design question of resolution against swath is a sensor-layer consequence that requires a width-versus-factor model this simulation does not have; it is reported as a limitation and future work, not as a demonstrated result. Fifth, the operational layer: the model covers single-pass scheduling decisions only — no dynamic task arrival, no downlink or

ground-station modeling, and no replanning loop — and no wall-clock runtime figure is registered for the promoted runs, so the platform is offered as a falsifiable scheduling substrate, not as an operationally timed planner; operational-cost evaluation is future work (Section 8).

7.2. Azimuth ambiguity versus the literature: a direction contrast

The registered correction behind this section is blunt: in this model the ambiguity-versus-squint direction is the *opposite* of the literature anchor it would most naturally be cited against. Under the production PRF and the registered calibration grid, the sliding-spotlight ambiguity ratio improves monotonically with squint: from -21.35 dB at zero squint to -23.32 dB at the 26° calibration-grid edge (the registered SLS cap is 26.9°). The stripmap ratio likewise improves (-21.35 to -23.32 dB over the same registered 0–26° calibration span), and the spotlight ratio is squint-invariant at -7.05 dB, exactly as the degenerate-integral structure of Section 3.3 predicts. That the stripmap and sliding-spotlight calibration sequences read identical values is not a transcription error: with $\rho_{a,\mathrm{SLS}} = A\,D/2$ the sliding-spotlight Doppler bandwidth is $B_{D,\mathrm{SLS}} = v_s \cos\bar{\psi}/(A\,D/2)$, and once the spectrum is compressed by the sliding factor A its bandwidth equals the stripmap bandwidth $2\,v_s \cos\bar{\psi}/D$ for every A, so under the same pattern G and the same PRF the two sequences degenerate point by point. Song et al.'s platform, by contrast, degrades with scan angle (the anchors quoted in Section 2) (Song et al., 2025); those values are cited as that platform's literature readings only. The contrast also carries its model-dependence: it is a consequence of adopting the rotation-centre semantics (under which the compression factor is squint-independent) together with the registered omission of the absolute Doppler centre, and no experiment in this paper discriminates the two steering semantics; the literature-side mechanism — a degrading compression factor competing with bandwidth narrowing — is not modeled here, so the comparison establishes a difference of registered model behavior, not a measurement that refutes the anchor. A scope note on the PRF: the production PRF is a registered platform constant of this world,

and no PRF-selection or band-widening criterion is exercised; the SLS and SPT processing bandwidths at zero squint, 15.2 and 25.3 kHz, are several times the PRF, and their aliasing structure enters the admission readings through the AASR model evaluated at the registered PRF rather than being avoided by a PRF choice.

The mechanism is the Doppler bandwidth. In this model the spectral compression factor is the design constant $a_{\mathrm{res}} \equiv A$ at every squint (Section 3.1), so the *only* squint-dependent channel into the ambiguity integral is the Doppler bandwidth $B_D = v_s \cos\bar{\psi}/\rho_a(m,0)$: squint narrows B_D, a narrower band aliases less of the ambiguity power, and the ratio improves. The registered direction contrast therefore differs by mechanism, not merely by sign: it is what remains once the pseudo-boundary of the linearised steering law — under which the compression factor itself would collapse with squint — is retired as the extrapolation artifact it is (Section 3.1). We flag the contrast rather than reconcile it: the two statements are made under different steering semantics, and the comparison is only legitimate with that stated. A caliber note on the spotlight reading: -7.05 dB is the library-edge constant evaluated at the platform-level scene extent (5 km, worst edge), whereas admission evaluates each task at its own scene extent through the 11-point in-window rule of Section 3.3: the spotlight aliasing term scales with scene size over range. The two figures are therefore different calibers of the same model, not a contradiction; the admission readings are the ones every feasibility statement inherits.

7.3. Model caveats

f_{ac} absolute position is omitted. The ambiguity folding is integrated about the direct-current (DC)-relative coordinate, so the absolute Doppler centre f_{ac} never enters the model; only the bandwidth and the compression factor depend on squint. The measured magnitude of the omission at the design-domain edge (worst side, 45°, an extrapolated-domain reading) is 345.7 kHz (95.2 times the production PRF of 3630 Hz, and 300.6 times the 1150 Hz registration key), so the folded ambiguity structure sits far outside the band and the omission is

defensible; it is disclosed regardless, and pushing the platform to its full 45° design domain must carry this number with it. A second simplification of the same family: the antenna pattern carries azimuth-frequency dependence only — there is no cross-range (elevation-plane) pattern factor — so the absolute ambiguity level is an azimuth-integration reading; no verdict in this paper rests on the unmodeled elevation-plane attenuation, and the direction contrast of Section 7.2 is carried by the bandwidth channel alone. Because the omitted term is a two-way attenuation applied to the ambiguous returns and not to the beam-centre principal image, including it could only lower the modelled ratio: the absolute ambiguity level reported here is an upper bound, and admission statements inherit that direction. Its magnitude is not quantified here.

The C2 maneuver angle is a two-sided lower bound. The inter-task requirement of Section 3.2 forms the maneuver angle from the difference of the tasks' start squints. This is a lower bound on the true attitude excursion from two independent sides: it omits the beam sweep during each dwell (the per-task sweep rate is 0.6686°/s at 30° incidence, 0.5599°/s at 45°, and 0.4233°/s at 60°, i.e. a 1.7–2.7° excursion across a 4 s dwell that the start-point difference never sees), and it understates the off-vertical attitude compression by 15–25%. The 25% inter-task pad is therefore a *margin argument*, not a correctness argument: it buys slack against a bound that is systematically low, and the paper never describes the bound as an exact measure of the attitude change. One cross-reference closes this caveat: the per-task sweep rates quoted here are the antenna steering rates of Section 3.2's definition, and the antenna's own sweep during a dwell is the quantity the 1.7–2.7° excursion describes — the dwell that feeds C2 comes from the broadside-derived closure whose registered approximation status Section 3.2 records.

g_T has no external cross-validation. The transmit-path gain term is self-derived and internally verified; the registered search for an external source found no noise-equivalent equation set in the Song platform paper (zero hits across its full text) and the textbook is not in the library with resolvable pages (Song

et al., 2025; Cumming and Wong, 2005). The verification route is therefore the internal identity chain of Section 3.3 and its regression guard, and the absence of external cross-validation stands as a limitation in its own right. Consistent with the objective-level scope note of Section 3.3, no *calibrated* radiometric quantity enters the objective; the gain term shapes the resolution index and the uncalibrated relative range factor carries the range dependence, but no noise model is imported.

The range-resolution anchor is a design constant. ρ_{rg} is held at the design-anchor constant of 0.5 m, registered at 45° incidence with scene-centre semantics. Under the real-window incidence band (30–60°) a physical range resolution would vary by roughly $-18\%/+41\%$ across the band (the $\sin\eta_i$ projection of the frozen design-incidence bandwidth: -18% at 60°, $+41\%$ at 30°, each relative to the 45° anchor); the constant is a modeling choice that keeps the quality readings comparable across windows, and the magnitude narrative of any quality reading inherits this caveat.

The quality index depends on the scheduled window. The range factor $(R_0/R)^3$ of Section 3.3 is evaluated at the slant range of the window in which the selected start falls, and a task's windows are enumerated at different transits, hence at different slant ranges. The index of a task is therefore not a property of the task alone: moving a selection to a different transit changes its index, and with a frozen mode assignment that change is one of the quantities a schedule can still choose. The variation is not corrected anywhere in this paper, and it is the same non-normalisation that leaves index levels incomparable across families (Section 5.2).

The caliber that travels with the reading. The fulfillment ratio is computed on the library caliber, and its limitation is stated with it: it measures a task's realised value against its own zero-squint optimum over its own windows — a within-task, within-family quantity. It is not a calibrated radiometric statement, it does not compare across families (the units expand scene-dependently,

Section 5.2), and it is not the delivered resolution of the schedule.

The maneuver fallback and the transfer-time composition. The production transfer bound is the preregistered fallback: the two-phase maneuver model underpredicts the measured Kim RMT it is calibrated against (0.000 s at the calibration point against a 4.07 s measurement), so every inter-task transition carries the measured 4.07 s bound instead of the model value. Replayed over the 1,577 consecutive pairs of the registered schedules, the model value sits below the bound in 84% of pairs (median 2.409 s, range 0.021–6.88 s); the pairs above the bound are the large-angle transitions whose shortfall the 25% inter-task pad is the declared margin for. The median transfer requirement decomposes as 5.643 s of pad-carrying dwell, 2.784 s of maneuver bound and the 2 s settle; pad and settle enter linearly, so halving the pad moves the median requirement by roughly 5% — a sensitivity stated here rather than swept. The scene-velocity convention (inertial, Section 3.1) scales the dwell linearly, and the coverage readings compare schedules inside one convention, so they are invariant to it.

Per-mode calibers are not economically comparable. The spotlight admission in the mode library is defined per point target while the sliding-spotlight scenes are scene-level, and the scheduler consumes the scene-level dwell; the two calibers are registered as asymmetric. Any statement that trades dwell economy across modes — a spotlight dwell against a sliding-spotlight dwell as if they purchased the same thing — is outside what the model supports, and none is made.

7.4. Design semantics and pre-answers

Why not jointly optimize mode and time?. Because the mode is a product-requirement property, not a scheduler preference. The mode follows from the ordered (resolution requirement, scene size) pair by the coarsest-sufficient library table; making it a decision gene would let the optimizer silently upgrade or downgrade the product the customer requires. The sliding factor is likewise a system parameter. The evidence this paper can offer for the joint formulation being unnecessary here is narrower than that argument needs, and stating it

precisely matters: the coverage response is a construction response (Section 4.5), so in this scenario world a unit's outcome is set before the search runs, and a mode gene would add decision variables to a search whose measured contribution is two units out of 260; whether a mode gene would change the served set under a different demand mix is untested. Joint mode-and-time allocation remains future work for regimes where that activation exists.

The reading is single-budget. The curve of Section 6 is read at one registered budget, and a future budget extension can move it: nothing in the registered design forbids a re-run at a larger budget, and if a larger budget changed the served sets, the registered criteria of Section 5.3 would be re-measured rather than assumed. In practice the budget was fixed by a registered decision — raising it was judged to buy a different search rather than a different construction, since the construction face carries the response (Section 4.5) — so the extension is available on paper and priced out by policy. That asymmetry is disclosed rather than hidden, and it bounds every reading in this paper.

8. Conclusion

This paper put the required resolution into the scheduling constraint set and measured what tightening it costs. On the platform leg, the imaging mode is resolved from the product requirement and its parameters follow the requirement per task, so the delivered resolution equals the requirement at the selected start by construction; the dwell is then an endogenous consequence of the mode, the requirement and the selected start, and the maneuver-coupled feasibility chain prices every schedule in physically meaningful time. The quality index states a platform capability and supplies a reported reading — a fulfillment ratio on the library caliber — and it carries no selection pressure: coverage is the only objective, and a unit's front is a tie-set on it.

The result is a coverage–demand curve over a frozen nine-point grid, read under a fixed mode assignment. Tightening is where the coverage moves: the 100-drawn / 99-admitted family drops 26.0 percentage points from the as-placed

requirement to $s = 0.9$ and 46.6 points to the tightest feasible point, the single-satellite control drops 46.0 points at $s = 0.9$ and has no feasible individual at all at the tightest point — an infeasibility record, not a coverage of zero — and the 50-drawn / 46-admitted family holds within 0.27 points of its as-placed coverage at every tighter point. Loosening recovers less: 4.15 percentage points for the 100-task family against its own saturation and 2.36 points for the 50-task family (0.93 paired per seed). The fixed-dwell counterfactual quantifies what the endogenous dwell buys: planned with the library dwell, the dense families violate the transfer constraint in 20–29% of consecutive pairs and land 18–30 percentage points short. Two of the three frozen shape criteria fail on the 50-task family, and both failures are reported as this world's answer; the tight-arm penalty of the previous world is withdrawn rather than quietly carried forward. The anchor already attains the unit maximum in 258 of 260 units — which is what makes the response a construction result rather than a search finding.

The claims stay inside those bounds. They are requirement sensitivity under a frozen mode assignment, at one registered budget, with the capacity law stated in caliber and without a number. The sensor-layer question of resolution against swath, the joint mode-and-time formulation, the recomputation of the serial bound under this scenario world, and an operational-cost evaluation are future work rather than results.

CRediT authorship contribution statement

Liu Shuhao: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Data availability

The promoted experiment artifacts (scenario manifests, per-unit pilot records across the three budget tiers — the pre-scan convergence-calibration pilots, after which this world reports the single budget of 40/60 (Section 5.3) — and the convergence-calibration registration), the preregistration file with its dated amendment chain, the caliber registry, and the verification script that recomputes the registered verdicts from those artifacts will be deposited on Zenodo with a persistent identifier upon acceptance of the paper; during peer review they are available from the corresponding author on request. No proprietary data were used; the scenario world is generated from a registered synthetic template.

References

Caleiras, M., Moniz, S., Nascimento, P.J., 2026. A Constraint Programming Model for the Super-Agile Earth Observation Satellite Imaging Scheduling Problem, in: ICORES 2026, SciTePress, pp. 110–121. doi:10.5220/0014407000004055.

Chang, Z., Zhou, Z., Xing, L., Yao, F., 2021. Integrated scheduling problem for earth observation satellites based on three modeling frameworks: an adaptive bi-objective memetic algorithm. Memetic Computing 13, 203–226. doi:10.1007/s12293-021-00333-w.

Chen, M., Du, Y., Tang, K., Xing, L., Chen, Y., Chen, Y., 2024. Learning to Construct a Solution for the Agile Satellite Scheduling Problem With Time-Dependent Transition Times. IEEE Transactions on Systems, Man, and Cybernetics: Systems 54, 5949–5963. doi:10.1109/TSMC.2024.3411640.

Cumming, I.G., Wong, F.H., 2005. Digital Processing of Synthetic Aperture Radar Data: Algorithms and Implementation. Artech House.

Deb, K., Agrawal, S., Pratap, A., Meyarivan, T., 2002. A fast and elitist multiobjective genetic algorithm: NSGA-II. IEEE Transactions on Evolutionary Computation 6, 182–197. doi:10.1109/4235.996017.

Du, Y., Xing, L., Li, L., 2025. Satellite Scheduling Engine: General-Purpose Modeling and Intelligent Optimization Methods. De Gruyter. doi:10.1515/9783111537191.

Gawiejnowicz, S., Kurc, W., 2015. Structural properties of time-dependent scheduling problems with the lp norm objective. Omega 57, 196–202. doi:10.1016/j.omega.2015.04.015.

Gawiejnowicz, S., Kurc, W., Pankowska, L., 2009. Equivalent time-dependent scheduling problems. European Journal of Operational Research 196, 919–929. doi:10.1016/j.ejor.2008.04.040.

Guo, D.M., Xu, H.P., Li, J.W., 2011. Extended Wavenumber Domain Algorithm for Highly Squinted Sliding Spotlight SAR Data Processing. Progress In Electromagnetics Research 114, 17–32. doi:10.2528/PIER11012010.

Habet, D., Vasquez, M., Vimont, Y., 2010. Bounding the optimum for the problem of scheduling the photographs of an Agile Earth Observing Satellite. Computational Optimization and Applications 47, 307–333. doi:10.1007/s10589-008-9220-7.

He, L., Liu, X., Laporte, G., Chen, Y., Chen, Y., 2018. An improved adaptive large neighborhood search algorithm for multiple agile satellites scheduling. Computers & Operations Research 100, 12–25. doi:10.1016/j.cor.2018.06.020.

He, L., de Weerdt, M., Yorke-Smith, N., 2019. Time/sequence-dependent scheduling: the design and evaluation of a general purpose tabu-based adaptive large neighbourhood search algorithm. Journal of Intelligent Manufacturing 31, 1051–1078. doi:10.1007/s10845-019-01518-4.

Kim, H., Chang, Y.K., 2020. Optimal mission scheduling for hybrid synthetic aperture radar satellite constellation based on weighting factors. Aerospace Science and Technology 107, 106287. doi:10.1016/j.ast.2020.106287.

Kim, H., Park, J., Chang, Y.K., Lee, S.H., 2021. Optimal Attitude Maneuvering Analyses for Imaging at Squint Staring and Sliding Spotlight Modes of SAR Satellite. Aerospace 8, 277. doi:10.3390/aerospace8100277.

Kim, H.D., Seong, J.D., 2019. Multi-objective Heuristic Design Approach for SAR Mission for Monitoring Local Target Area. International Journal of Aeronautical and Space Sciences 20, 518–524. doi:10.1007/s42405-018-0129-9.

Li, J., Zhu, J., Xu, D., Wang, J., Li, Z., Zhang, K., 2024. Earth Observation Satellite Scheduling With Interval-Varying Profits. IEEE Transactions on Aerospace and Electronic Systems 60, 8273–8288. doi:10.1109/TAES.2024.3427090.

Li, L., Feng, Q., Li, K., Liang, Y., 2026. Efficient Generation and Quality Screening of Visible Windows for Regional SAR Reconnaissance. doi:10.48550/arXiv.2604.19539. arXiv preprint.

Liu, W., Zhang, D., Tian, Y., Shao, X., 2025. A weighted multi-objective hierarchical adaptive scheduling method for SAR satellite regional observation mission. Aerospace Systems doi:10.1007/s42401-025-00354-0.

Long, Y., Shan, C., Shang, W., Li, J., Wang, Y., 2024. Deep Reinforcement Learning-Based Approach With Varying-Scale Generalization for the Earth Observation Satellite Scheduling Problem Considering Resource Consumptions and Supplements. IEEE Transactions on Aerospace and Electronic Systems 60, 2572–2585. doi:10.1109/TAES.2024.3379169.

Mittermayer, J., Lord, R., Börner, E., 2003. Sliding Spotlight SAR Processing for TerraSAR-X using a New Formulation of the Extended Chirp Scaling

Algorithm, in: IGARSS 2003 (IEEE Int. Geoscience and Remote Sensing Symposium), Toulouse, pp. 1462–1464.

Peng, G., Dewil, R., Verbeeck, C., Gunawan, A., Xing, L., Vansteenwegen, P., 2019. Agile earth observation satellite scheduling: An orienteering problem with time-dependent profits and travel times. Computers & Operations Research 111, 193–205. doi:10.1016/j.cor.2019.05.030.

Prats, P., Scheiber, R., Mittermayer, J., Meta, A., Moreira, A., 2010. Processing of Sliding Spotlight and TOPS SAR Data Using Baseband Azimuth Scaling. IEEE Transactions on Geoscience and Remote Sensing 48, 770–780. doi:10.1109/TGRS.2009.2027701.

Qin, J., Bai, X., Du, G., Liu, J., Peng, N., Xu, M., 2024. Multi-satellite Scheduling for Moving Targets Using the Enhanced Hybrid Genetic Simulated Annealing Algorithm and Observation Strip Selection. IEEE Transactions on Aerospace and Electronic Systems 60, 5773–5800. doi:10.1109/TAES.2024.3397958.

Ren, M., Zhang, H., Yu, W., 2025. High-Resolution High-Squint Large-Scene Spaceborne Sliding Spotlight SAR Processing via Joint 2D Time and Frequency Domain Resampling. Remote Sensing 17, 163. doi:10.3390/rs17010163.

Shen, X., Lu, Z., Li, L., Chen, Y., Li, X., Wang, J., Yao, W., 2024. Multi-strip Stitching Imaging Mission Planning Method for SAR Satellite Regional Mapping Considering Onboard Energy Consumption. IEEE Transactions on Geoscience and Remote Sensing doi:10.1109/TGRS.2024.3513483.

Song, Y., Bi, H., Cai, F., Li, G., Zhang, J., Hong, W., 2025. Spaceborne Sparse SAR Imaging Mode Design: From Theory to Implementation. Sensors 25, 3888. doi:10.3390/s25133888.

Wang, B., Feng, Y., Zhang, G., Zhang, L., Yang, Y., 2024a. Memetic multi-objective discrete jaya algorithm for cooperative scheduling of multiple agile

earth observation satellites. IEEE Transactions on Aerospace and Electronic Systems 60, 8086–8099. doi:10.1109/TAES.2024.3429302.

Wang, X., Wang, S., Meng, F., 2024b. Research of multi-satellite mission planning method based on multi-chromosome genetic algorithm, in: 2024 IEEE International Conference on Control Science and Systems Engineering (ICCSSE), pp. 328–332. doi:10.1109/ICCSSE63803.2024.10823951.

Xin, S., Mengdao, X., Jinsong, Z., Huitao..., L., 2024. SAR multi-satellite collaborative complex area observation planning based on improved genetic algorithm. National Remote Sensing Bulletin doi:10.11834/jrs.20243258@.

Zhang, Q., Xiao, F., Ding, Z., Ke, M., Zeng, T., 2018. Sliding Spotlight Mode Imaging with GF-3 Spaceborne SAR Sensor. Sensors 18, 43. doi:10.3390/s18010043.

Zhou, G., Jin, Z., Liu, D., 2025. Deep Reinforcement Learning-Based Two-Phase Hybrid Optimization for Scheduling Agile Earth Observation Satellites. Remote Sensing 17, 1972. doi:10.3390/rs17121972.